Escalation Chain — QA Test Cases

Task 3.17 · v0.1 · Owner: Syed Sarim Jawad Zaidi · Based on docs/api escalation, medication, wellness, panic and emergency_contact contracts (dev, 19 Oct 2026)

Purpose: prove that an unanswered alert climbs Primary → Secondary → External contact (SMS), that opening it stops the chain, and that Panic reaches everyone. These cases are run live at ICP-3 against Abdullah's backend and Talha's app. Results are recorded on a copy of the QA Checklist template (docs/qa/).

1. Test setup

1.1 Test accounts (create once, before the run)

Role	Name in tests	Set-up
Dependent	Ammi	Registered, KYC verified (any CNIC not ending in 0), linked to Ali.
Primary Guardian	Ali	Phone + CNIC verified. Sets schedules, emergency contact, permissions.
Secondary A	Sara	Invited by Ali, verified. Permission emergency_alerts = ON (medication_management = ON for E-06/E-07 checks).
Secondary B	Bilal	Invited by Ali, verified. ALL permissions OFF (proves Panic ignores permissions and normal alerts respect them).
External contact	Uncle Bilal	PUT /dependents/{id}/emergency-contact by Ali, e.g. 0300 7654321 (removed for E-18).

1.2 Start the backend with short timers (so a full chain takes ~4 minutes, not 40)

In backend\ with the virtual environment active, PowerShell:

```
$env:SCHEDULER_ENABLED="true"
$env:MEDICATION_WINDOW_S="60"            # dose counts as missed 1 min after its time (normal: 30 min)
$env:ESCALATION_PRIMARY_TIMEOUT_S="60"   # Primary has 1 min (normal: 5 min)
$env:ESCALATION_SECONDARY_TIMEOUT_S="60" # Secondary has 1 min more (normal: 5 min)
uvicorn app.main:app --reload
```

The server job runs once a minute, so every step can take up to one extra minute. Times use Karachi time (Asia/Karachi).

1.3 Where to look

- **Each person's alerts:** GET /notifications (logged in as that person)
- **Chain level:** GET /dependents/{id}/escalations → tier (primary / secondary / external_contact) and next_escalation_at
- **SMS:** the server terminal prints "MOCK SMS -> +92…" (the gateway is mocked; no real SMS is sent)
- **In the app (once Talha's 3.14 is built):** Guardian Alerts screen (mockup 08) and Missed dose screen (08b)

2. Test cases

A. Missed medicine dose (FR-D1, FR-D5, FR-R1)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-01	FR-D1	Medicine schedule for Ammi due in 2 min. Short timers on.	1. Do not confirm the dose. 2. Wait until the window ends (+1 min for the job). 3. Read GET /notifications as Ali, Sara, Bilal and Ammi.	Ali: one critical missed_dose, data.tier = primary. Sara, Bilal: nothing yet. Ammi: no extra alert (only the reminder + due alert she already had). GET /dependents/{id}/medication-doses: status = missed.		
E-02	FR-D5	E-01 alert open, nobody opens it.	1. Wait the Primary timeout (+1 min). 2. GET /dependents/{id}/escalations. 3. Read notifications as Sara and Bilal.	Event tier = secondary. Sara (emergency_alerts ON): critical missed_dose, tier = secondary. Bilal (permission OFF): nothing.		
E-03	FR-D5, FR-R1	E-02 still unopened. Emergency contact set.	1. Wait the Secondary timeout (+1 min). 2. Check escalations and the server log. 3. Read Ali's notifications.	Tier = external_contact; next_escalation_at = null. Server log: exactly ONE "MOCK SMS -> +92…" with the missed-dose wording. Ali: standard external_contact, result = sent, "Emergency contact messaged".		
E-04	FR-M2	New missed-dose alert at Level 1.	Ali opens it: POST /notifications/{id}/acknowledge (or /escalations/{event_id}/acknowledge). Wait 2 timer periods.	Chain stops: tier stays primary, next_escalation_at = null. Sara gets nothing. No SMS.		
E-05	FR-M2	Alert reached Level 2.	Sara acknowledges. Wait one timer period.	Chain stops at secondary. No SMS. Repeating the acknowledge is safe (same reply).		
E-06	FR-D1	Missed dose with an open alert.	Ali: POST …/medication-doses/{id}/extend with {"minutes": 30}. Then repeat with {"minutes": 200}.	1st: dose back to pending, alert closes, standard all-clear to everyone told; Ammi gets critical medication_extended with the new time. 2nd: 422 VALIDATION_ERROR (5–120 only).		
E-07	FR-D1	Missed dose with an open alert at Level 1.	Ali: POST …/escalate. Then call it again.	1st: tier = secondary, notified = 1 (Sara). Secondary timer starts from now. 2nd: 409 ALREADY_ESCALATED.		
E-08	FR-F1	Missed dose, alert open, no SMS yet.	Ammi confirms late (step 1 then step 2).	Dose = taken; alert closes; standard all-clear "… took … late" to everyone told. No SMS.		
E-09	FR-R1	Missed dose alert already reached SMS (E-03).	Ammi confirms late.	Alert closes; server log shows ONE all-clear SMS: "The alert for … is closed".		

B. Missed wellness check (FR-D4, FR-D5)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-10	FR-D4	Wellness check due in 2 min, grace 5 min. NO location fix in last 30 min.	1. Wait for the prompt (Ammi's notifications: wellness_check_prompt). 2. Do not answer. Wait grace + 1 min.	Check status = escalated. Ali: critical wellness_check, tier = primary.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-11	FR-D4.3	Same, but send a location fix for Ammi during the grace period.	Do not answer. Wait grace + 1 min.	Check closes as device_active. NO alert to anyone.		
E-12	FR-D4	E-10 alert open.	Ammi answers late: POST /wellness-checks/{response_id}/respond.	Alert closes; standard all-clear to everyone told. Answering twice is safe.		
E-13	FR-D5, FR-R1	E-10 alert, nobody opens it.	Wait both timers.	Primary → Secondary (Sara) → external_contact. One SMS with "not answered a wellness check".		

C. Panic (FR-D2)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-14	FR-D2	Bilal has NO permissions at all.	Ammi: POST /dependents/{id}/panic.	201, notified = 3. Ali, Sara AND Bilal each get critical "PANIC ALERT" (permissions ignored on purpose).		
E-15	FR-D2	Panic already open.	Ammi presses panic again.	200, already_active = true. No second round of notifications.		
E-16	FR-D5	Panic open, nobody opens it.	Wait the Primary timeout (+1 min).	Goes straight to external_contact (no secondary step: everyone was already told). One SMS: "HUMSAFAR URGENT: … pressed the panic button…".		
E-17	FR-D2	Panic open.	1. Sara: POST …/panic/resolve. 2. Ammi: POST …/panic/resolve. 3. Ammi resolves again.	1: 403 NOT_PRIMARY_GUARDIAN. 2: status = resolved; standard all-clear "Panic alert closed" to all told (+ all-clear SMS if one was sent). 3: 409 NO_ACTIVE_PANIC.		

D. External contact, skipped levels, privacy and security

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-18	FR-R1	NO emergency contact set.	Let a missed-dose alert reach the last level.	No SMS. Ali gets ONE critical "No emergency contact set". Then Ali adds a contact while the alert is still open → SMS goes out on the next run.		
E-19	FR-R1	Contact set; SMS gateway failing.	*[How to make the mock gateway fail: ask Abdullah]* Let an alert reach the last level.	Retried every 60 s, up to 3 tries. Then Ali gets critical "Emergency contact could not be reached".		
E-20	FR-D5	Sara's emergency_alerts switched OFF (no allowed Secondary).	Let a missed-dose alert go unopened.	Secondary level is skipped (nobody to tell); next stop is external_contact.		
E-21	FR-R1	Sara (Secondary) and Ammi (Dependent).	PUT /dependents/{id}/emergency-contact as Sara, then as Ammi. Then a bad phone "12345" as Ali.	Sara: 403 NOT_PRIMARY_GUARDIAN. Ammi: 403 FORBIDDEN. Ali bad phone: 422 VALIDATION_ERROR. Good phone saved as +92…		
E-22	Privacy	Any alert about Ammi.	Read Ammi's GET /notifications during E-01 – E-13.	Ammi is never told about her own missed-dose / wellness escalation.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
E-23	Security	A Guardian NOT linked to Ammi.	POST /escalations/{event_id}/acknowledge for Ammi's event.	404 ESCALATION_NOT_FOUND (cannot see or stop someone else's alert).		

3. ICP-3 live run (one sequence, about 15 minutes)

The roadmap's ICP-3 demo: Panic, a missed wellness check all the way to the external contact, and a medicine confirmation — on the Guardian dashboard. Run with the short timers from 1.2.

Step	Who	Action	Everyone should see
1	Ali	Open the Guardian app home (mockup 07). Check the emergency contact is set.	Status tiles, no open alerts.
2	Ammi	Dependent home → PANIC → hold (mockups 20, 21).	Ammi: "Help is coming" (21b). Ali, Sara, Bilal: PANIC ALERT at once.
3	Ali	Tap the panic banner → "I've seen it".	Chain stops; no SMS. Then Ammi taps "I am safe now" → all-clear to everyone.
4	Ammi	Wellness check prompt arrives (22). Do NOT answer. No location fixes.	After grace + 1 min: Ali alerted (Level 1).
5	Nobody	Do not open the alert. Wait ~2 min.	Sara alerted (Level 2), then ONE MOCK SMS in the server log (Level 3); Ali told "Emergency contact messaged".
6	Ammi	Answer the wellness check late ("Yes, I am OK").	Alert closes; all-clear to everyone; all-clear SMS in the log.
7	Ammi	Medicine due → "I took it" → "Yes, confirm" (23, 23b).	Dose = taken; no alert. Guardian schedule shows it taken.

Pass rule: every step behaves as written, no Critical/High bug open. Record results and bugs on a copy of the QA Checklist (save as Humsafar_QA_ICP3_Results_v1.0).

4. Open points

- E-19 needs a way to make the mock SMS gateway fail on purpose — to confirm with Abdullah.
- Timers are the same for every Dependent (5 + 5 min); a per-Dependent setting is an open decision (sds_clarifications items 54–57). Cases assume the current behaviour.
- App-side checks (sound, screens) depend on Talha's 3.10–3.14; until then run the cases through Postman.